

KONNERUD, Norway

WMO: 014770

Lat: 59.7128N

Lon: 10.1458E

Elev: 193

StdP: 99.03

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.6	-15.9	-21.4	0.6	-18.2	-18.6	0.7	-14.5	6.8	1.3	5.4	-0.4	1.0	310	0.568

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	26.1	16.7	24.4	16.0	22.7	15.0	18.3	23.1	17.3	21.7	16.4	20.6	2.4	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.7	12.2	19.9	15.7	11.4	18.9	14.7	10.7	17.8	52.4	23.1	49.5	21.3	46.5	20.9	22.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5.4	4.6	3.8	-20.2	28.5	2.9	1.9	-22.3	29.9	-24.0	31.0	-25.7	32.1	-27.8	33.5
			-20.6	19.6	3.0	1.6	-22.7	20.7	-24.5	21.6	-26.2	22.5	-28.4	23.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.5	-5.0	-3.0	0.2	4.7	10.2	13.9	16.7	15.2	10.9	5.7	0.4	-4.0
	DBStd	8.42	5.08	5.08	4.11	3.27	3.93	2.57	2.65	2.60	2.77	3.26	4.11	5.63
	HDD10.0	2223	464	364	303	162	46	1	0	0	23	138	287	433
	HDD18.3	4692	723	598	562	409	251	135	64	104	224	393	537	691
	CDD10.0	597	0	0	0	3	53	119	209	161	48	4	0	0
	CDD18.3	24	0	0	0	0	1	2	15	6	0	0	0	0
	CDH23.3	263	0	0	0	0	20	34	151	57	1	0	0	0
	CDH26.7	31	0	0	0	0	2	4	21	4	0	0	0	0
(14) Wind	WSAvg	1.5	1.4	1.5	1.6	1.7	1.7	1.8	1.6	1.4	1.4	1.4	1.3	1.4
(15) Precipitation	PrecAvg	957	75	47	48	59	67	79	86	93	101	112	111	72
	PrecMax	1450	173	145	134	163	136	186	173	169	315	276	463	150
	PrecMin	703	8	7	0	8	10	10	8	20	37	28	20	19
	PrecStd	152	45	28	36	37	33	41	38	35	59	61	83	37
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.7	9.4	15.0	18.9	25.8	26.7	28.8	27.1	21.8	16.0	11.4	9.1
		MCWB	3.9	4.9	8.4	11.3	15.1	16.7	18.1	16.0	15.4	11.8	8.4	6.5
	2%	DB	5.0	7.0	11.3	16.0	22.8	24.1	26.7	25.1	19.6	13.8	9.7	6.7
		MCWB	2.1	3.2	5.7	9.4	13.7	15.1	17.2	16.9	13.5	10.3	6.9	4.8
	5%	DB	3.3	5.2	8.5	13.3	20.4	22.0	24.9	23.1	17.7	12.4	7.7	4.8
		MCWB	1.2	2.1	3.8	7.3	12.4	13.8	16.4	15.8	12.5	9.8	6.0	2.8
	10%	DB	1.8	3.7	6.3	11.2	17.9	20.1	23.1	21.1	16.0	11.2	6.2	3.4
		MCWB	0.4	1.3	2.4	6.1	11.3	12.7	15.6	14.9	11.9	8.9	4.6	1.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.6	5.6	8.6	12.4	16.7	17.9	20.0	19.1	16.7	13.2	9.5	7.0
		MCDB	6.4	8.7	14.1	17.5	22.2	24.0	25.3	23.6	19.7	15.0	10.5	8.6
	2%	WB	3.0	4.0	6.1	10.0	15.1	16.4	18.8	17.9	14.8	11.7	7.5	4.9
		MCDB	4.7	6.1	10.7	15.3	20.2	21.4	24.3	21.8	17.8	12.8	8.9	6.0
	5%	WB	1.6	2.7	4.5	8.3	13.6	15.2	17.8	16.9	13.8	10.6	6.2	3.3
		MCDB	2.7	4.5	7.6	12.1	18.7	19.8	22.5	20.7	16.3	11.9	7.4	4.3
	10%	WB	0.5	1.7	3.0	6.8	12.2	14.1	16.8	15.9	12.9	9.4	5.0	1.8
		MCDB	1.5	3.2	5.9	10.2	16.5	18.2	21.1	19.9	15.1	10.9	6.1	3.0
(35) Mean Daily Temperature Range	MDBR		5.9	7.4	9.0	9.8	10.1	9.6	10.1	10.1	8.8	7.2	5.3	6.0
	5% DB	MCDBR	7.6	9.1	12.3	14.2	14.7	13.3	13.5	12.7	12.4	9.5	8.2	7.4
		MCWBR	5.9	6.4	7.3	7.7	7.0	6.3	5.7	5.5	6.7	6.3	6.4	6.3
	5% WB	MCDBR	6.9	7.9	11.8	12.8	12.6	10.8	11.6	10.5	9.6	7.7	6.9	6.9
		MCWBR	5.8	6.0	7.6	7.6	6.6	6.3	5.9	5.8	6.2	6.1	6.2	6.2
(40) Clear-Sky Solar Irradiance	taub		0.251	0.291	0.315	0.356	0.354	0.345	0.365	0.358	0.328	0.315	0.271	0.225
	taud		2.229	2.316	2.357	2.355	2.440	2.484	2.445	2.458	2.533	2.492	2.316	2.135
	Ebn at Noon		556	726	822	845	875	889	859	837	810	700	529	436
	Edh at Noon		40	65	85	102	101	98	100	92	72	56	37	29
(44) All-Sky Solar Radiation	RadAvg		0.31	0.93	2.44	3.87	4.91	5.60	4.97	3.92	2.59	1.17	0.41	0.20
	RadStd		0.06	0.18	0.24	0.54	0.54	0.42	0.64	0.30	0.32	0.22	0.11	0.02

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (53 neighbors)		N/A	N/A	+1.56	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page